SatQuery AI | Remote Sensing Analysis Report

SIH 2026 Problem Statement SIH26167 • Automated Vision-Language Intelligence

Analysis ID:	70a3b99c-c928-4521-8fa3-2478a6715930	**Date/Time:**	2026-10-03 05:32:29
Detected Task:	**SINGLE_IMAGE_ANALYSIS**	**Pipeline Mode:**	**BASELINE / DEMO MODE**
User Query:	*"Calculate vegetation coverage and spectral indices for this region."*	**Confidence:**	**92%** (Calibrated Metric)

1. Executive Summary & Natural Language Answer

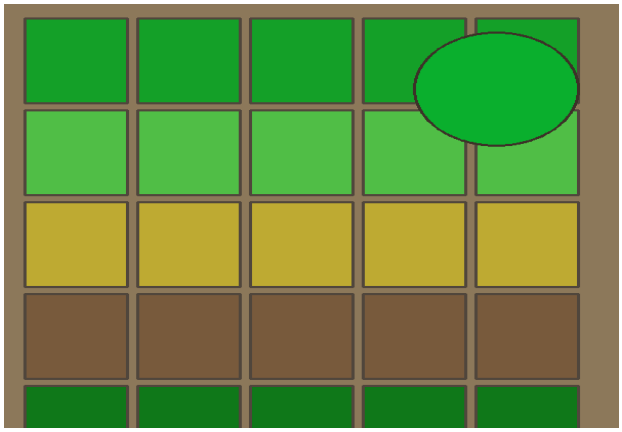
Single-image remote sensing analysis completed for the 600×600 scene. The image exhibits a mean radiometric brightness of 98.79 and an edge complexity of 4.51% (Moderate). Spectral vegetation analysis (VARI) indicates approximately 40.03% green biomass cover and 0.0% potential water/low-reflectance surface.

Key Analytical Findings:

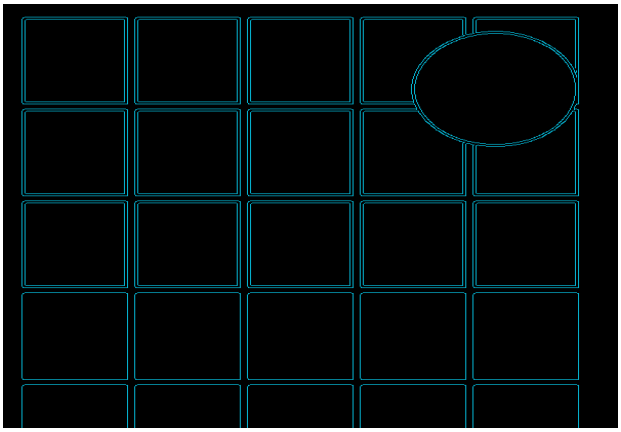
• Image dimensions: 600 × 600 pixels across 3 spectral channel(s).
• Mean scene radiometric brightness is 98.79 with std dev 53.01.
• Estimated green biomass / vegetation coverage: 40.03% of scene.
• Structural edge density: 4.51% (Moderate complexity).

2. Visual Evidence & Spatial Overlays

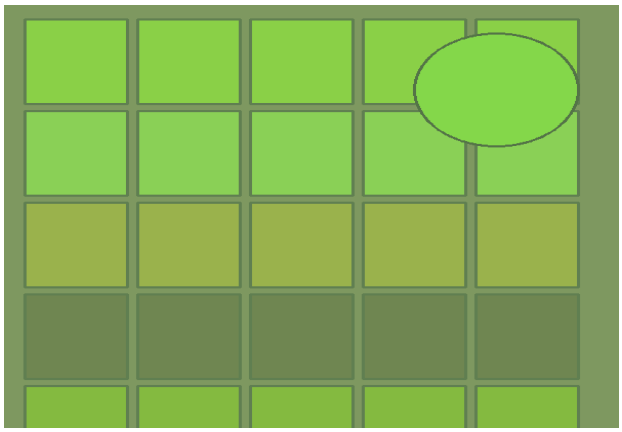
Input Satellite Raster (Primary)



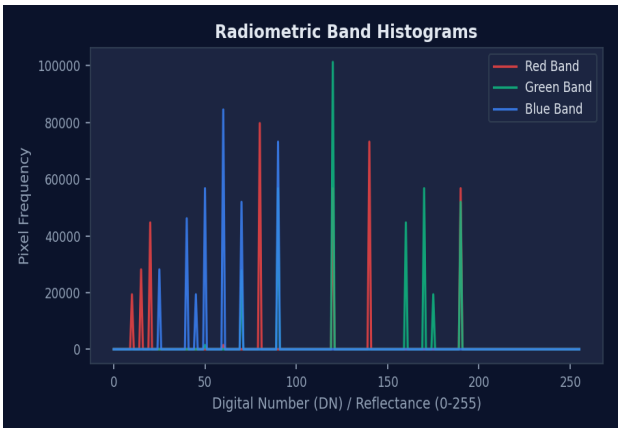
Structural Edge & Boundary Map



Visible Spectral Vegetation Index (VARI)



Radiometric Histogram



3. Agentic Execution Summary & Verification

Tools/Models Invoked:	RadiometricProfiler, VARI_IndexCalculator, SobelEdgeAnalyzer, HistogramEngine
Execution Time:	1.358 seconds
Hardware Engine:	CPU (NumPy/OpenCV Engine)
Scientific Rationale:	Computed directly from verified radiometric reflectance values, gradient operators, and standard spectral index formulas.

Notice: Generated automatically by SatQuery AI. In Baseline/Demo mode, metrics are derived from verified radiometric signals, Otsu thresholding, SSIM, and spectral formulas without synthetic hallucination.